WILSON MOSES
DATA SCIENCE × AI ENGINEERING
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DATA SCIENCE FIELD NOTES / 04B

HEALTHCONNECT PROJECT SUMMARY

Week 4 problem definition, data assessment and proposed next phase

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PROGRAMME	AnalystLab Africa - Data Science Internship
PROJECT	HealthConnect No-Show Prediction
SUBMISSION	30 August 2026 | Week 4 | Batch D

Educational fictional-clinic project. Not a clinical decision system or validated operational tool.

Project purpose

The Data Science track examined whether HealthConnect's fictional appointment history can support early prediction of no-shows. The intended future value is to help clinic staff prioritise suitable attendance support and improve use of scheduled capacity.

> Current status: business and analytical definition completed; initial data suitability assessed; model training and deployment not yet completed.

Verified snapshot

Measure	Result
Appointments	5,000
Variables	18
Anonymised patient IDs	1,696
No-shows	2,423
Attended	2,314
Cancelled	263
Binary cohort	4,737
No-show rate	51.15%

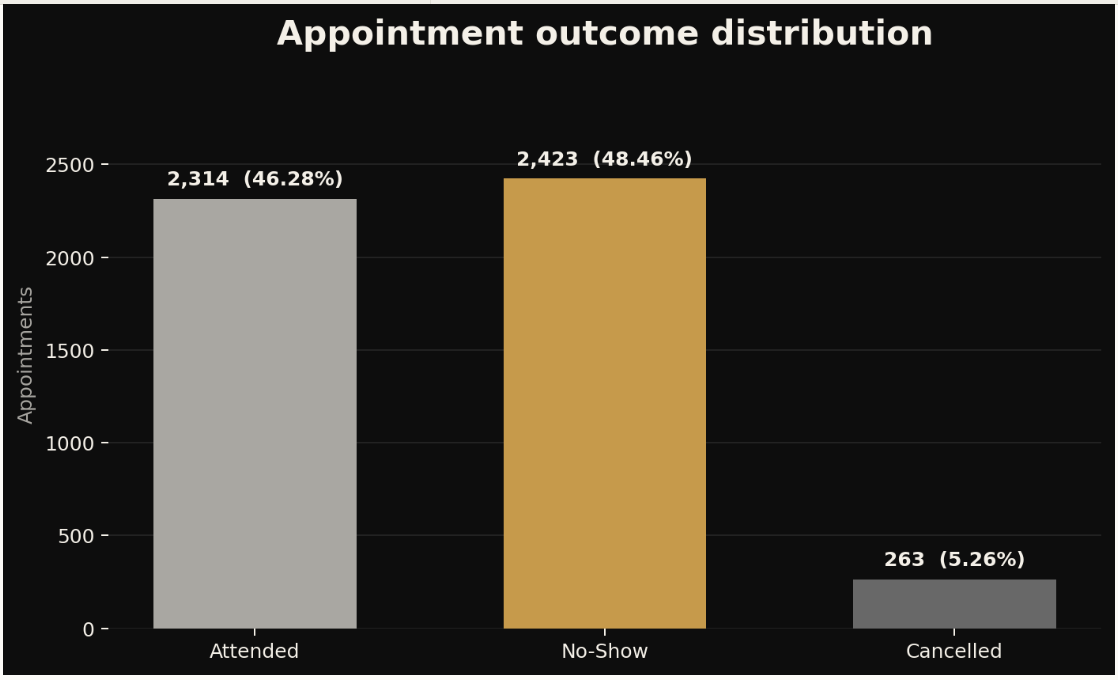

Figure 1. Supplied appointment outcomes.

Key observations

- No exact duplicate rows or duplicate appointment identifiers were detected.
- Distance is missing in 90 records and waiting time in 60 records.

- The 1,366 None reminder-channel values represent no reminder being sent.
- The observed date format differs from the data dictionary's stated format.
- There are 737 Sunday appointments despite the stated Sunday closure.
- Repeated patient IDs contain inconsistent age histories.

Proposed approach

The first proposed task is supervised binary classification, with No-Show encoded as 1 and Attended as 0. Cancellations remain preserved but temporarily excluded. Only information available before the prediction point should enter the model.

Stage	Proposed action
Baseline	Establish a simple reference prediction
Interpretable model	Train logistic regression after leakage-safe preprocessing
Comparison	Test selected tree-based alternatives where justified
Evaluation	Use class-sensitive metrics, calibration and subgroup error analysis
Operational review	Select thresholds using clinic priorities and staff capacity

Responsible-use boundaries

- The project is fictional and educational.
- Predictions must not diagnose patients, rank access to care or deny services.
- Associations must not be treated as explanations of behaviour.
- Any intervention should be supportive, transparent and reviewed by staff.
- Operational benefit must be tested rather than assumed.

Week 5 focus

- Conduct structured exploratory analysis.
- Investigate missingness, Sunday scheduling and repeated-ID inconsistencies.
- Confirm the prediction time and feature availability.
- Create a documented cohort and leakage-safe preparation plan.
- Define baseline, evaluation and fairness procedures before modelling.

Conclusion

HealthConnect is suitable for a responsible educational prototype, but the current evidence does not support operational claims. Progress should continue through documented quality resolution, leakage-safe preparation, careful evaluation and a clearly defined supportive action.